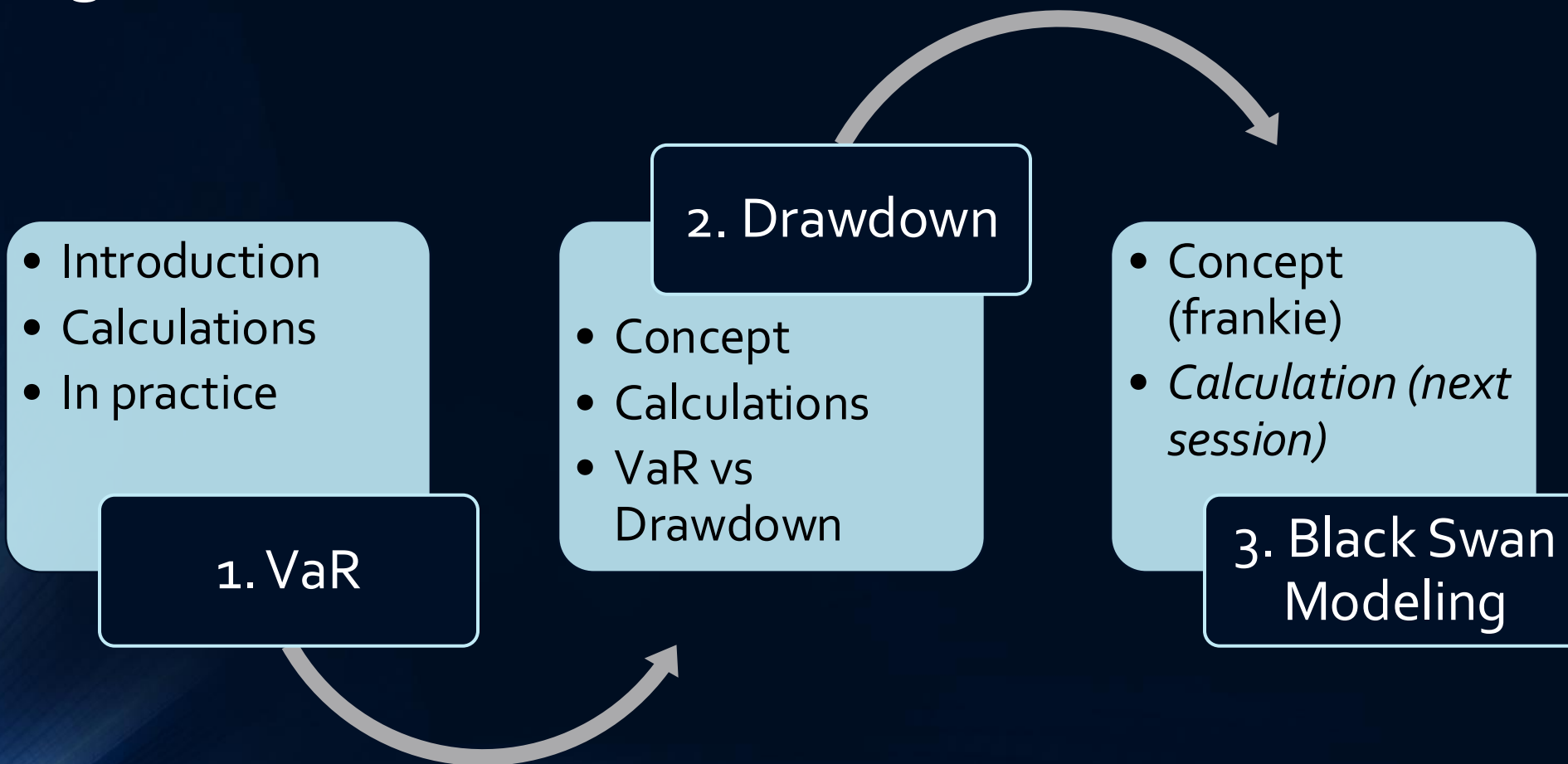


QMMML Week 7 ☺

QUANT TRADING: RISK TEAM (FRANKIE & ANYA)

Agenda



VaR (Value at Risk)

- You're in charge of managing a £20,000 investment portfolio !



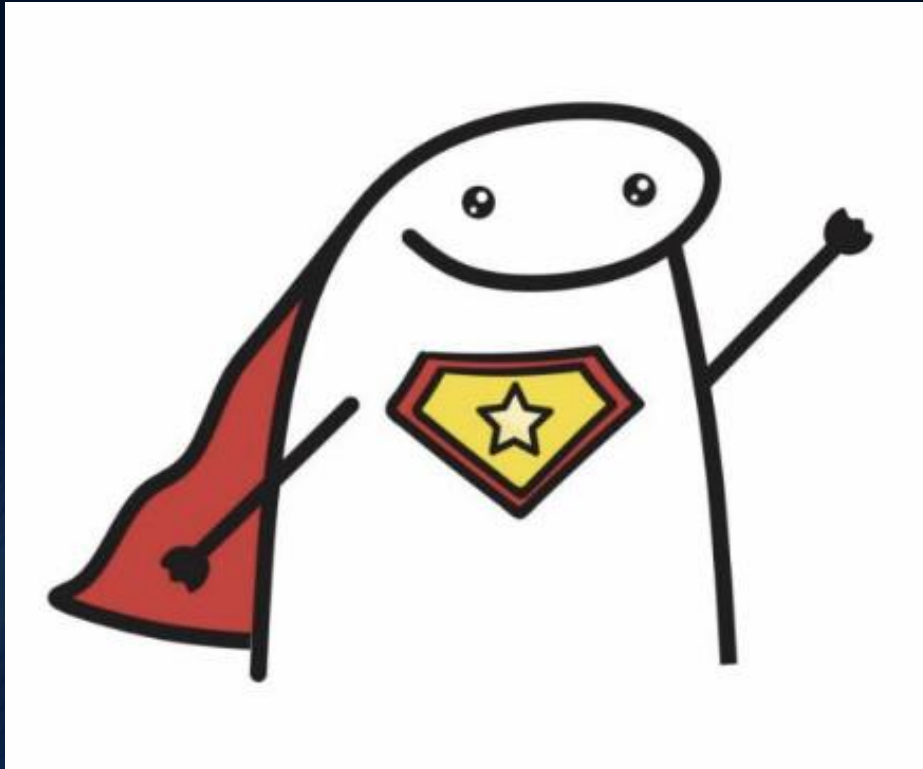
You 😊

VaR (Value at Risk)

- “What’s the most money I could lose in a bad day?”



VaR (Value at Risk) comes in !



VaR Estimates:

- How much you could lose,
- Over a period of time (1 day in our case),
- With a certain confidence level (usually 95% or 99%).

*any confidence level can be used but 95% and 99% are most typically used because they balance realism and risk sensitivity.

VaR (Value at Risk) comes in !

VaR Estimates



Example:

- If a portfolio has a 1-day 95% VaR of £1000
- It means, "there is a 95% chance you will lose no more than £1000 tomorrow...but 5% chance you could lose more."

VaR = worst expected loss under normal conditions.

Back to you:



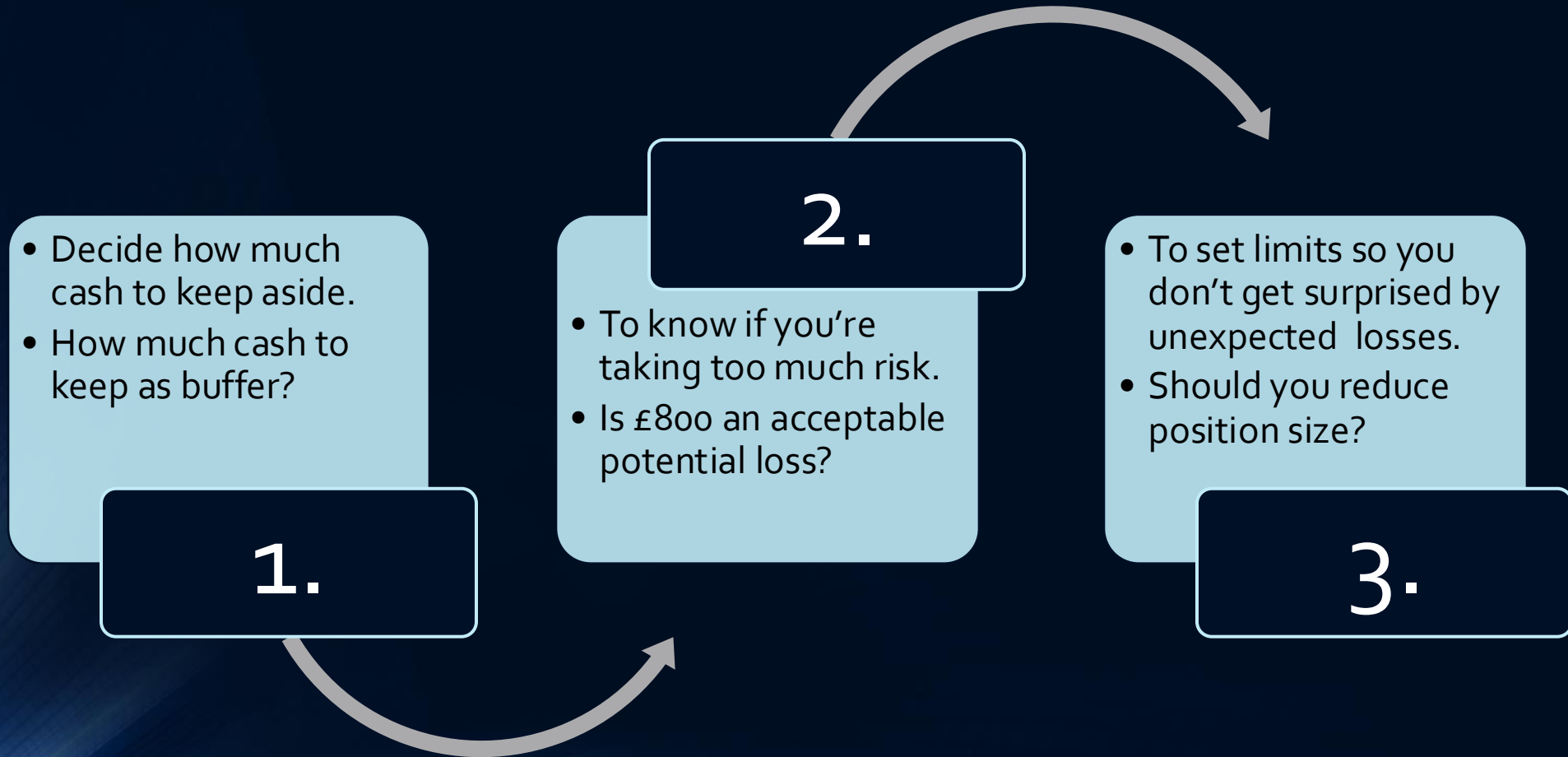
- You have a £20,000 stock portfolio.
- You want to understand your risk for the next day? (what is the most you are likely to lose in a normal bad day at 95% confidence level?).

Using Historical VaR Method:

- 95% VaR on £20,000 stock.
- Step 1: Gather past 10 daily returns (-4%,-3%,-2%,-1%,0%,+1%,+1.5%,+2%,+3%,+4%).
- Step 2: Sort them from worst to best (already sorted).
- Step 3: Pick 5th percentile return for 95% VaR
(5% of 10 = 0.5, hence -4% is the 95% historical VaR).
- Step 4: Convert the % lost in pounds;
$$\text{VaR} = 4\% * £20,000 = £800$$

Final Ans: Your 1-day 95% Historical VaR = £800

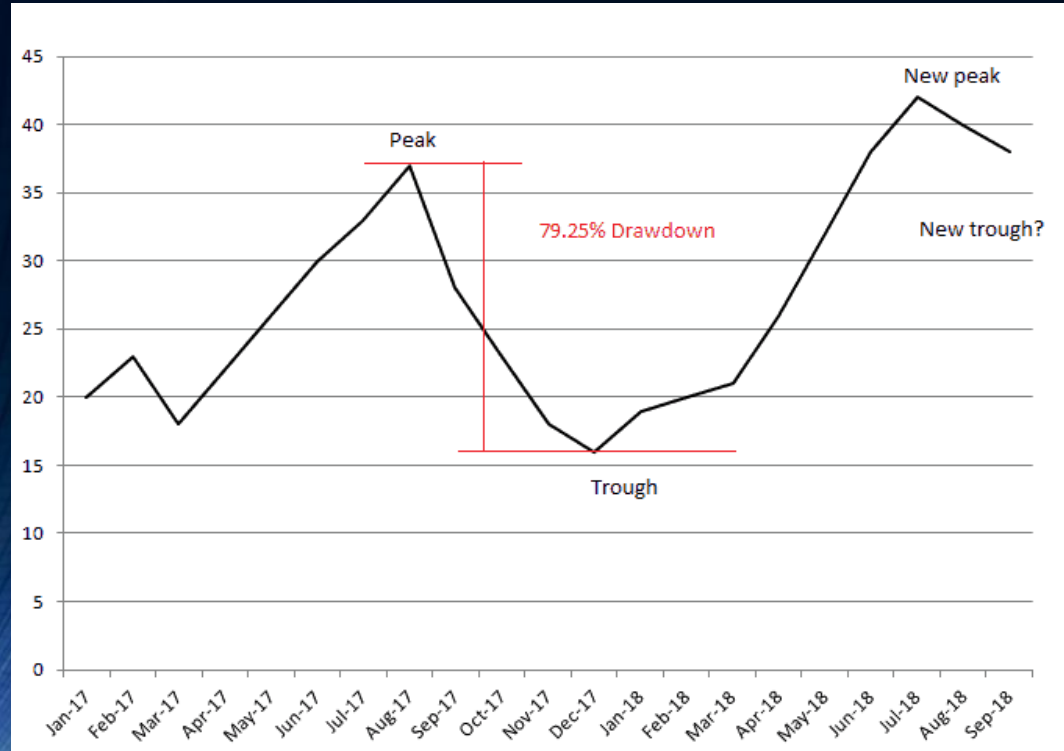
VaR-Why is it calculated?



Questions? :)

*there are 2 other methods used to calculate VaR: Variance/Covariance and Monte Carlo Simulation.

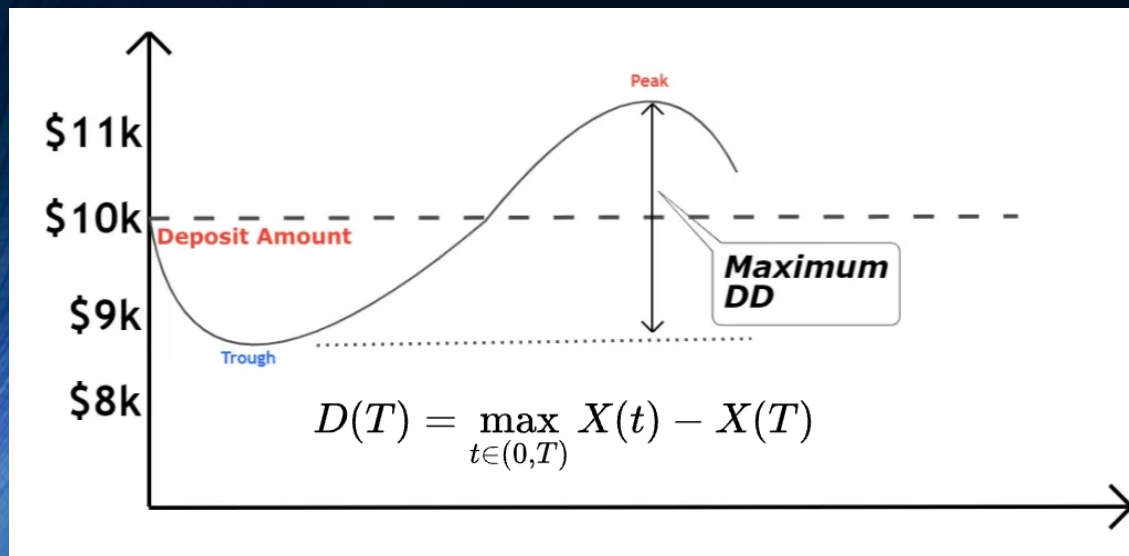
Draw Down metrics – Why



- You're considering a new trading strategy that:
 - Has historically been quite profitable.
 - Averages **over 20% annual return** on investment.
- However, historical performance also shows the strategy has experienced **drawdowns** of:
 - **\$6,000**
 - **\$7,000**
 - And on one occasion, even **\$10,000**.
- Despite these drawdowns, the strategy is **profitable overall**, with:
 - A **hypothetical \$5,000 investment** growing to **over \$25,000** in **four years**.
- **Why drawdowns still matter:**
 - Suppose you invest **\$5,000** and start trading right before a drawdown begins.
 - If the strategy hits a **\$6,000 drawdown** (the lowest previously recorded):
 - Your **entire investment would be wiped out**.
 - You would **not survive** long enough to benefit from the eventual gains.

Three Measures of Risk:

- Maximum drawdown (MaxDD)
- Average drawdown (AvDD)
- Maximum drawdown duration



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Three Measures of Risk:

- Maximum drawdown (MaxDD)
- Average drawdown (AvDD)
- Maximum drawdown duration

The **average drawdown** (AvDD) up to time T is the time average of drawdowns that have occurred up to time T :

$$\text{AvDD}(T) = \frac{1}{T} \int_0^T D(t) dt$$

Three Measures of Risk:

- Maximum drawdown (MaxDD)
- Average drawdown (AvDD)
- Maximum drawdown duration

Date	Portfolio Value	Running Max	Drawdown
Jan	100	100	0.00
Feb	95	100	-5%
Mar	90	100	-10%
Apr	85	100	-15% ▼
May	95	100	-5%
Jun	105	105	0.00

Jan → Jun is the maximum drawdown period

The **maximum drawdown period** is the **longest duration** for the maximum drawdown.

Let's say you have a time series of portfolio values P_t over time $t = 1, 2, \dots, T$:

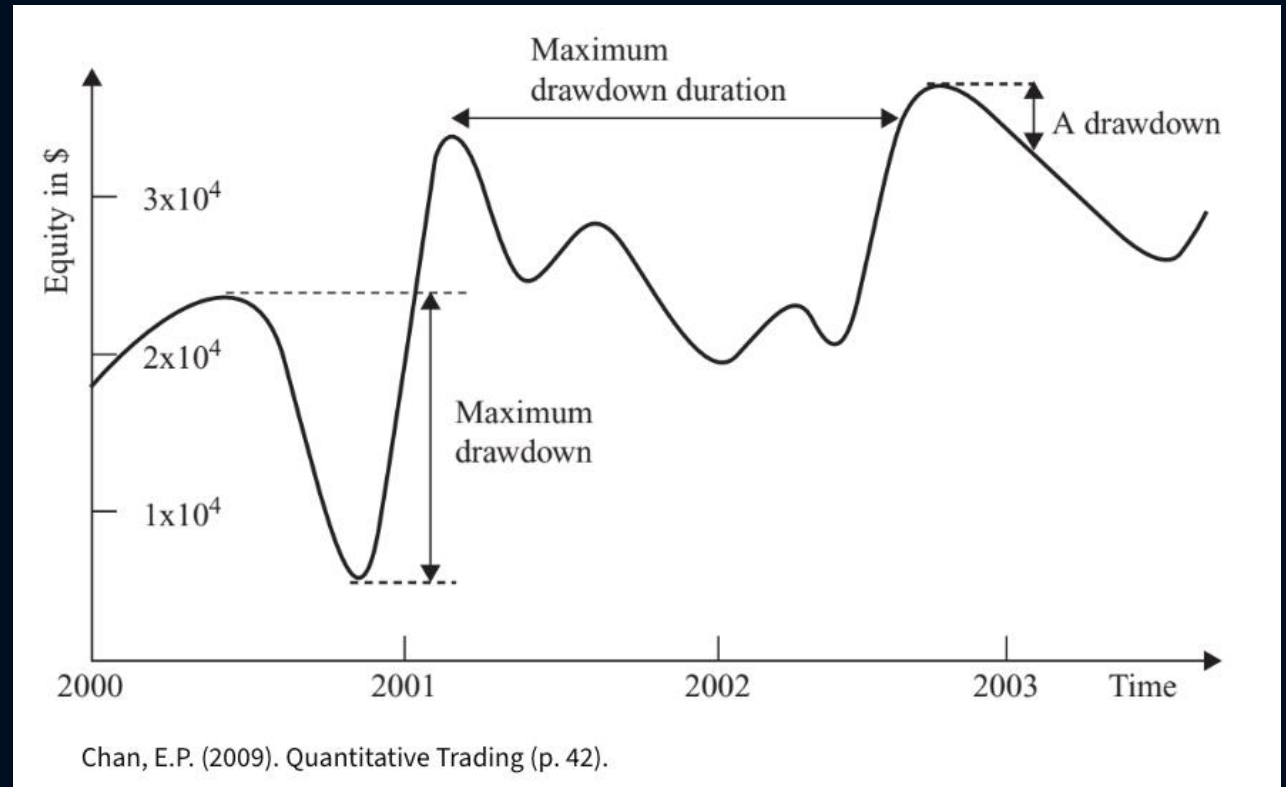
- **Find the running maximum** up to each point:
- **Compute drawdown at each point:**

$$D_t = \frac{P_t - M_t}{M_t}$$

- **Find the time t^* where D_t is minimal** (i.e., most negative) — this is the **max drawdown depth**.
- **Trace back to the time t_{peak} when the most recent max M_t occurred** before t^* .
- **The drawdown period is $t^* - t_{peak}$.**
- **Continue forward to the next time the portfolio value exceeds M_t —that's the recovery date, and the full drawdown period ends there.**

Three Measures of Risk:

- Maximum drawdown (MaxDD)
- Average drawdown (AvDD)
- Maximum drawdown duration

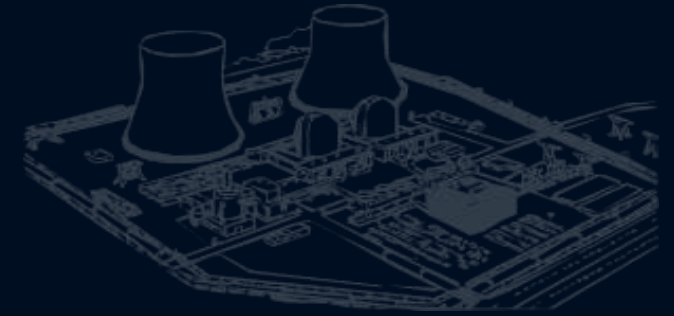


Concept	Value at Risk (VaR)	Maximum Drawdown (MDD)
What it measures	Maximum potential loss over a given time horizon at a specific confidence level	Largest observed peak-to-trough loss during a period
Type	Probabilistic, predictive	Historical, realized
Input	Distribution of returns or historical data	Time series of portfolio or asset values
Output	A dollar value or percentage loss (e.g., "You won't lose more than \$X with 95% confidence over 1 day")	A single worst-case percentage or dollar decline
Assumptions	Often assumes normal distribution (unless otherwise modeled)	No assumptions — it uses actual historical data
Captures extreme loss?	No — VaR stops at a confidence threshold (e.g., 95% or 99%)	Yes — includes all observed losses
Black Swan sensitivity	Poor — misses tail risks unless modeled using methods like CVaR or EVT	Good — if such events happened, they are captured
Commonly used by	Banks, risk managers, regulators	Hedge funds, portfolio managers
Decision support	Helps determine required capital reserves	Helps assess strategy durability and risk tolerance

Conclusions

- Losing client's accounts is equivalent to death of business;
- Highly unlikely to hold an account which was in a drawdown for 2 years;
- Highly unlikely to be permitted to have a 50% drawdown;
- Shutdown condition: 20% drawdown;
- Warning condition: 15% drawdown;
- Longest time to get out of a drawdown - 1 year.

Black swans modelling





A turkey is fed for 1000 days.

Each passing day confirms to its statistics department that the human race cares about its welfare “with increased statistical significance”.



A turkey is fed for 1000 days.

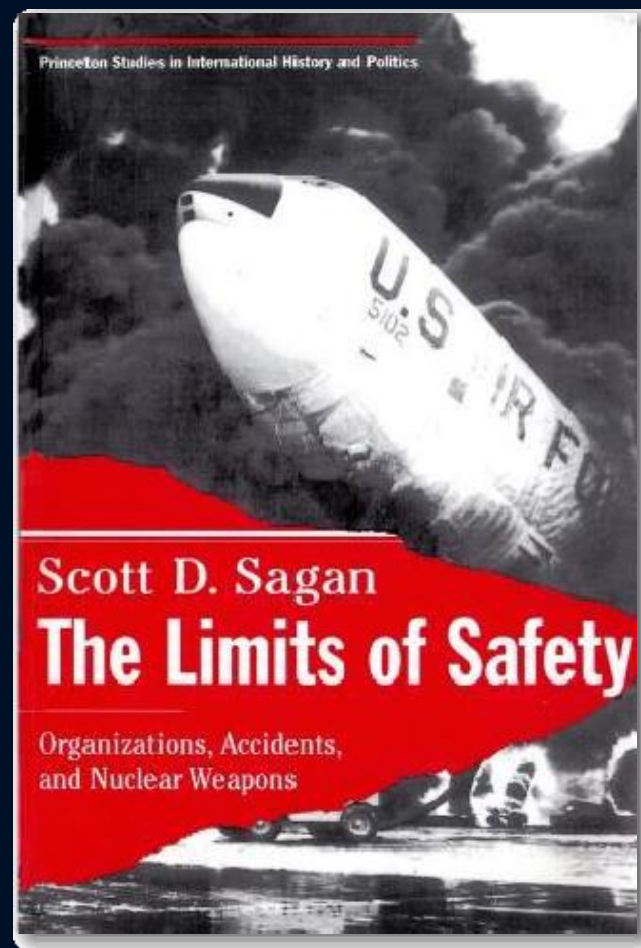
Each passing day confirms to its statistics department that the human race cares about its welfare “with increased statistical significance”.

On the 1001st day, a little before Thanksgiving, the turkey has a surprise.

“*Things that have never happened before, happen all the time.*”

—*Scott D. Sagan, The Limits of Safety: Organizations, Accidents, And Nuclear Weapons, 1993*

In risk analysis, these are called “unexampled events” or “outliers” or “black swans”.



The term “black swan” was used in 16th century discussions of impossibility (all swans known to Europeans at the time were white).

Explorers arriving in Australia discovered a species of swan that is black.

The term is now used to refer to events that occur though they had been thought to be impossible.



Black swans

- ▷ Characteristics of a black swan event:
 - an **outlier**
 - lies outside the realm of regular expectations
 - nothing in the past can convincingly point to its possibility
 - carries an **extreme impact**
- ▷ Note that:
 - in spite of its outlier status, it is often easy to produce an explanation for the event *after the fact*
 - a black swan event may be a surprise for some, but not for others; it's a subjective, knowledge-dependent notion
 - warnings about the event may have been ignored because of strong personal and organizational resistance to changing beliefs and procedures